```
<!DOCTYPE html>
        if(river[y][WIDTH-1]) {
            salinity[y][WIDTH-1] = 1.0;
        }
    }

    // 上流から淡水
    for(let y=0; y<HEIGHT; y++) {
        if(river[y][0]) {
            salinity[y][0] *= 0.2;
        }
    }

    // 更新
    for(let y=1; y<HEIGHT-1; y++) {
        for(let x=1; x<WIDTH-1; x++) {

            if(river[y][x]) {

                const diffusion = (
                    salinity[y+1][x] +
                    salinity[y-1][x] +
                    salinity[y][x+1] +
                    salinity[y][x-1]
                ) / 4;

                const flow = salinity[y][x-1];

                newSalinity[y][x] = (
                    0.72 * salinity[y][x] +
                    0.18 * diffusion +
                    0.10 * flow
                );
            }
        }
    }

    salinity = newSalinity;
}

function draw() {

    ctx.fillStyle = 'black';
    ctx.fillRect(0,0,canvas.width,canvas.height);

    for(let y=0; y<HEIGHT; y++) {
        for(let x=0; x<WIDTH; x++) {
```

```
            if(river[y][x]) {

                const s = salinity[y][x];

                const r = Math.floor(255 * s);
                const b = Math.floor(255 * (1-s));

                ctx.fillStyle = `rgb(${r},0,${b})`;
                ctx.fillRect(x*CELL, y*CELL, CELL, CELL);
            }
        }
    }
}

function animate() {
    update();
    draw();
    requestAnimationFrame(animate);
}

createRiver(24);
animate();

</script>

</body>
</html>
```